

# Orchestrating a Controllable Engineering of Dual-Model Carbon Structure in Si/C Anodes

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Silicon-carbon composites (Si/C) with multistage structures enable structural integrity during cycling. However, the lack of controllable structure preparation on a large scale hinders the stability improvement in practical applications. Herein, a new strategy is proposed to synthesize kilogram-scale Si/C (PySi/C) featuring a dual-model carbon structure in one step. The controllable combination of an onion-like carbon coating on the Si surface with independent pyrolytic carbon is accomplished through the precise adjustment of the pyrolysis temperature. The dual-model carbon formation mechanism is unraveled, detailing the cooperative coupling of the nucleation laws of carbon compositions as well as the changes trends in morphology and crystallinity. This density functional theory and finite element analysis highlight the dual-model structure's essential contribution to the electrochemical behavior and structural stability. As expected, PySi/Cs anodes deliver stable cycling performance with retention of 91.5% after 800 cycles at 2 A g<sup>-1</sup>. Its comprehensive electrochemical performance surpasses that of the state-of-the-art kilogram-scale Si-based anode reported. Moreover, the assembled pouch cell exhibits actual competitiveness, showing a capacity of 1.97 Ah and a retention of 88.9% after 300 cycles at 1 C. This work provides valuable design concepts to further advance the development of Si/C anodes.

## 1. Introduction

Lithium-ion batteries with high energy density are being developed to cater to the rising energy demand of mobile devices and electric vehicles.<sup>[1,2]</sup> Silicon (Si) is promised a flourishing anode material thanks to its high capacity (3579 mAh g<sup>-1</sup>), low potential ( $\approx 0.2$  V vs Li<sup>+</sup>/Li), and abundant resources.<sup>[3]</sup> However, Si anodes experience a multiphase alloying reaction with lithium ions (Li<sup>+</sup>), which causes a nearly linear increase in volume expansion accompanied by the formation of an unstable solid electrolyte interphase (SEI) film.<sup>[4–6]</sup> Over time, this leads to gradual flocculation of Si, ultimately resulting in pulverization and detachment from the collector. In addition, Si still suffers from deficient electric conduction.

Because of their high strength, flexibility, electrical conductivity, and good compatibility, carbon materials are widely available for combining with Si to solve the above drawbacks simultaneously.<sup>[7–9]</sup> Essentially, the electrochemical performances of silicon-

carbon composites (Si/C) are decisively influenced by the carbon component.<sup>[10]</sup> Carbon-coated structures were initially believed to provide Si/C stabilization properties by alleviating swelling and improving SEI films. However, Si/C particle contacts are susceptible to severe multi-directional stresses during cycling, resulting in structural failure.<sup>[11–13]</sup> Therefore, introducing additional carbon structures is imperative to synergistically interact with the carbon coating, establishing a dual-model carbon structure.<sup>[14]</sup> In this case, additional carbon ameliorates the stress environment of Si/C particles, rapidly disperses stress, and provides a conductive carbon network within the system. Regrettably, existing methods for synthesizing Si/C anodes incorporating dual-model carbon present notable challenges, especially when endeavoring to scale up production to the kilogram scale.

Some representative physical approaches, such as ball milling,<sup>[15–17]</sup> spray drying,<sup>[18–20]</sup> and wet methods,<sup>[21,22]</sup> enable the preparation of scalable Si/C with multistage structures. However, ensuring the homogeneity of the chemical environment surrounding the Si in the system is difficult throughout these productions, thereby tending to the occurrence of stress

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concentration in specific regions. Alternatively, chemical vapor deposition (CVD) is a one-step method that allows for the controllable carbon protection on Si.<sup>[23–25]</sup> Unfortunately, the conventional CVD method is limited to depositing amorphous carbon layers solely on the surface of silicon nanoparticles (Si NPs). The loose carbon coating is a relatively weak binding force to Si, making the structure susceptible to damage during lithiation. The research found that by depositing carbon layers of different thicknesses on Si NPs, the capacity retention only increased from 35% to 61%.<sup>[25]</sup> Furthermore, the CVD method requires strict flow control of the gas-phase carbon source,<sup>[26,27]</sup> which seriously affects the preparation efficiency. Therefore, obtaining scalable Si/C with multistage carbon structures through facile methods is particularly crucial for practical applications.

Herein, we propose a strategy for the simultaneous synthesis of two structural carbon materials, guided by the principles of carbon deposition and nucleation. Building upon this concept, we design an original pyrolysis device to achieve the one-step preparation of kilogram-scale pyrolytic Si/C (PySi/Cs) with a dual-model carbon structure. This includes carbon coating on the Si surface and auto-nucleated pyrolytic carbon. The formation mechanism is corroborated through characterization analysis. The dual-model structure is also controllably adjusted by altering the pyrolysis condition, in terms of its content, morphology, and crystallinity. We also validated the positive effects of dual-model carbon in the electrochemical behavior. As a result, PySi/Cs exhibit superior electrochemical performances, with a retention of 91.5% after 800 cycles and rapid lithiation kinetics. The assembled pouch full battery demonstrates competitiveness with a capacity of 1.49 Ah and a retention of 88.9% after 300 cycles at 1 C. This work provides a scalable and effective approach to developing Si/C with a multistage protective structure, which contributes to the development of high-performance energy storage devices.

## 2. Results and Discussion

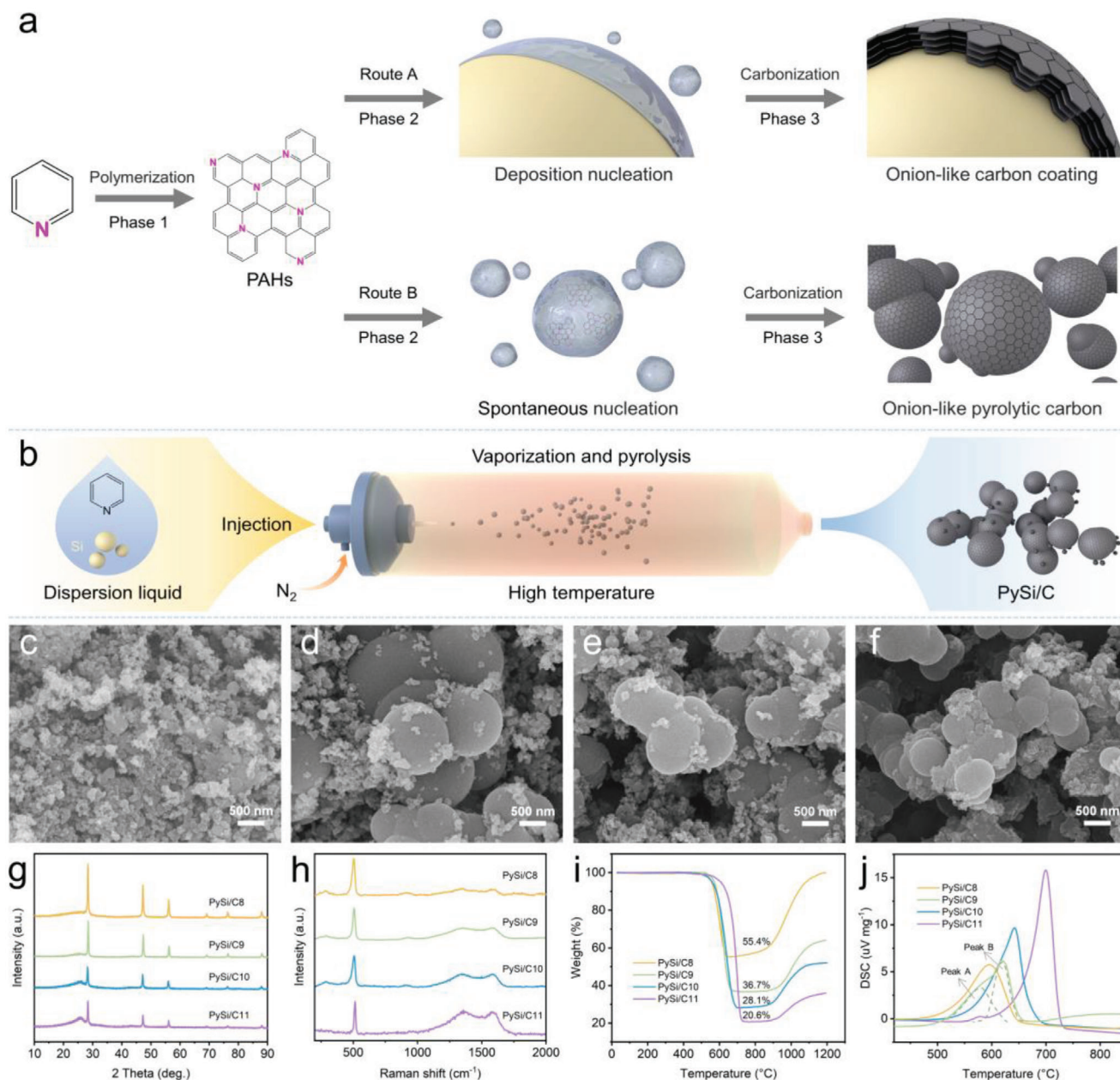
Initially, we devised a one-step synthesis strategy for two structural carbon components relying on the gas-phase carbonization mechanism. This strategy involves two-part synthetic routes (Figure 1a), which differs significantly from conventional CVD. In the first phase, the carbon source is converted into polycyclic aromatic hydrocarbons (PAHs) through pyrolysis, condensation, and polymerization reactions in the gas phase ( $C_5H_5N \rightarrow C_{5+i}H_{5+j}N_{1+k}$ ).<sup>[28–30]</sup> Some of the gas-phase carbon source is adsorbed on the external solid surface (Si) due to the presence of surface adsorption energy (route A),<sup>[31,32]</sup> while another part remains free in the system (route B). In the second phase, the partial pressure of the PAHs rises until it approaches the saturated vapor pressure as the gas-phase reaction progresses. In this case, PAHs will also begin to condense to form a liquid nucleus called gas-phase nucleation (a process known as the initial nucleation of carbon black).<sup>[33]</sup> Surface adsorption drives gas molecules to preferentially nucleate on the Si surface (route A). Subsequently, carbon molecules grow on the Si surface, ultimately forming the carbon-encapsulated Si structure ( $C_{5+i}H_{5+j}N_{1+k} \rightarrow C_mN_n + \gamma H_2$ ). In addition, supersaturation and temperature also trigger free PAHs self-nucleation (route B). In the final phase, the pyrolysis of the macromolecular liquid nucleus produces solid carbon par-

ticles. The carbon coating on Si will collaborate with independent pyrolytic carbon particles to construct a dual-model structure.

Based on the envisaged synthesis strategy, it is a prerequisite for the pyrolysis process to provide a sufficient gas-phase carbon source environment and a high temperature. Therefore, the liquid-phase carbon source is employed, which both enhances the dispersion of Si NPs and renders the reaction process more manageable. More importantly, the liquid-phase carbon source is instantly vaporized under high temperature to bring high pressure in the furnace, which in turn satisfies the condition of deposition nucleation. The carbon source used here is pyridine, whose high carbonization yield tends to form more carbon components. We designed scalable injection pyrolysis equipment and achieved kilogram-scale preparation (Figure S1, Supporting Information). Figure 1b illustrates the preparation process of PySi/C. PySi/Cs were synthesized directly by injecting Si-pyridine dispersion into a high-temperature furnace, and they were distinguished based on the pyrolysis temperature (for example, PySi/C8 is obtained at 800 °C).

Scanning electron microscopy (SEM) images show the morphologies of PySi/Cs. Only small particles with the same size, about 100 nm, as the Si NPs (Figure S2, Supporting Information) are observed in the viewing field of PySi/C8 (Figure 1c; Figure S3, Supporting Information). The difference is that when the temperature is raised to 900 °C and above, there are large particle structures in addition to small particles (Figure 1d–f; Figure S4–S6, Supporting Information), and their size decreases with increasing temperature. The distribution of elements in PySi/Cs particles is characterized by energy dispersive X-ray spectroscopy (EDS) to identify the element composition of different particles (Figure S3d–S6d, Supporting Information). The small particles are identified as Si/C particles, deriving from the fact that they contain both Si and C elements. In contrast, there is almost no Si element on the large particle sites, and we deduce that they are pyrolytic carbon derived from free carbon sources. The morphology results are consistent with our design concept, yet additional validation is required to substantiate the reliability.

The change in carbon component in different PySi/Cs is supported by X-ray diffraction (XRD) patterns and Raman spectra. The characteristic peaks evident in PySi/Cs correspond to the diamond cubic phase of Si- and broad peaks at about 26° correspond to carbon (Figure 1g).<sup>[34,35]</sup> As the pyrolysis temperature increases, the intensity of the broad peak gradually increases, indicating the formation of more carbon components. The typical Si peaks ( $520\text{ cm}^{-1}$ ) and carbon peaks ( $1342$  and  $1585\text{ cm}^{-1}$  for the D peak and the G peak, respectively) in all PySi/Cs are shown in Raman spectra (Figure 1h). From thermo-gravimetric analysis (TGA, Figure 1i), the Si contents in PySi/C8, PySi/C9, PySi/C10, and PySi/C11 are estimated to be 55.4, 36.7, 28.1, and 20.6%, respectively. Obvious weight increases of PySi/Cs occur at temperatures of 900–1200 °C, which is caused by the high-temperature oxidation of Si in the air.<sup>[2]</sup> Differential scanning calorimetry (DSC) curves deliver obvious exothermic peaks at temperatures between 500 and 800 °C (Figure 1j), according to the loss of carbon content. The exothermic peak area increases and the peak position gradually shifts blue, suggesting that the high temperature enhances both the content and thermal stability of the carbon component in PySi/Cs.<sup>[36]</sup> Focusing on the



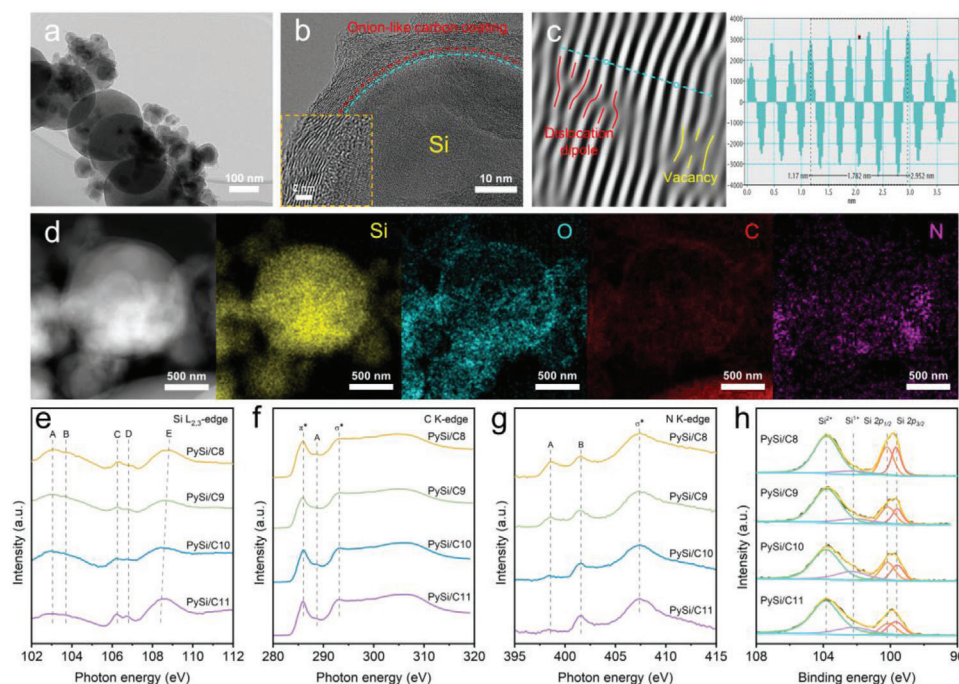
**Figure 1.** a) Illustration of the formation mechanism of carbon coating on Si NPs and pyrolytic carbon. b) Preparation illustration of PySi/Cs. SEM images of c) PySi/C8, d) PySi/C9, e) PySi/C10, and f) PySi/C11. g) XRD patterns, h) Raman spectra, i) TG curves, and j) DSC curves of PySi/Cs.

peak of PySi/C9 reveals two distinct peaks (dotted lines), signifying the existence of two carbon components with different crystallinities, a feature absent in PySi/8. This phenomenon occurs because the crystallinity of carbon grown on the substrate surface surpasses that of carbon grown by self-nucleation under identical conditions.<sup>[37,38]</sup> Similarly, the peaks of PySi/C10 and PySi/C11 consist of two peaks, but they are sharper, indicating that both carbons have high crystallinity.

To verify the above inference, namely, the large particles are independent pyrolytic carbon. We analyzed large particles isolated from PySi/Cs. Figure S7a–c (Supporting Information) show the SEM images of large particles from PySi/C9, PySi/C10,

and PySi/C11, respectively, which are consistent with those in Figure 1d–f. More importantly, no Si characteristic peaks found in XRD patterns identify they are only pyrolytic carbon with a higher crystallinity (Figure S7d, Supporting Information). The above results confirm the correctness of the synthesis strategy. At lower pyrolysis temperatures, carbon source molecules need to adsorb on the Si surface to reduce energy, ultimately forming a coating structure. Whereas at high temperatures, the carbon source molecules are driven by temperature to self-nucleate in addition to adsorbing on the Si surface. These two carbon components in PySi/Cs synergistically form a dual-model structure for Si NPs. What's more, the number of initial liquid nuclei





**Figure 2.** a, b) TEM and HRTEM images of PySi/C11. c) Inverse fast Fourier transform image of onion-like buffer coating in selected region of image and calculation of lattice spacing. d) EDS element mappings of PySi/C11. e) Si  $L_{2,3}$ -edge XANES spectra, f) C K-edge XANES spectra, g) N K-edge XANES spectra, and h) high-resolution Si 2p XPS spectra of PySi/Cs.

increases with the supersaturation and temperature.<sup>[39]</sup> This mechanism explains the gradual increase in pyrolytic carbon content and decrease in particle size at high temperatures. We also prepare PySi/C at 850 °C and 950 °C, both of which also consist of Si/C and pyrolytic carbon (Figure S8, Supporting Information). Comparing the morphology of PySi/Cs under different conditions, the pattern is consistent with the formation mechanism proposed above. Thus, varying the pyrolysis temperature enables the controllable adjustment of the dual-model structure. This approach also demonstrates its universality by allowing it to cover Si in any shapes, like flakes and flowers (Figure S9, Supporting Information). We have also attempted other carbon sources (toluene and n-heptane) and reconfirmed the feasibility of the method (Figure S10 and Note S1 in Supporting Information). In addition, the presence of pyrolytic carbon endows PySi/C with a higher tap density, with the density of PySi/C9, PySi/C10, and PySi/C11 being 0.65, 0.69, and 0.43 g cm<sup>-3</sup>, respectively (Figure S11 and Note S2, Supporting Information).

Transmission electron microscopy (TEM) and high-resolution TEM (HRTEM) are used to analyze structures of PySi/Cs in more detail. Focusing on small Si/C particles, this one-step injection pyrolysis allows for in situ onion-like graphitic carbon coating of Si NPs, which also demonstrates the presence of a deposition process (Figure 2a,b; Figure S12, Supporting Information). The graphitization of the onion-like carbon buffer structure increases with pyrolysis temperature rise, as shown by the reduction of carbon defects (Figure 2c; Figure S13, Supporting Information).<sup>[40]</sup> Similarly, the interplanar spacing reduces from 0.416 nm in PySi/C8 to 0.356 nm in PySi/C11, approaching the graphite (002) interplanar spacing of 0.34 nm (Figure S14, Supporting Information). Pyrolytic carbon has the same onion-like structure (Figure

S15, Supporting Information). The reason for this unique carbon structure is due to the layer-by-layer deposition of carbon molecules in pyrolysis. Subsequently, during the holding process, the carbon atoms rearranged and the lattice fringe spacing changed.<sup>[28]</sup> During the cycling, graphitic carbon layers will exert clamping forces through an interlayer sliding process to effectively alleviate the volume expansion of Si, which preserves structural integrity and enhances cyclic stability.<sup>[24]</sup> The elemental distribution of PySi/C is investigated again, using PySi/C11 as an example (Figure 2d; Figure S16, Supporting Information). Si is distributed in the particle's center; O is distributed in the middle layer; and C is distributed on the exterior. The presence of a silicon oxide layer in the middle (the region between the red dashed line and the blue dotted line in Figure 2b) permits the carbon layer to forge a stronger covalent bond with Si, while also acting as a barrier against the formation of an electrochemically inert silicon carbide layer.<sup>[24,41,42]</sup> The derived carbon is doped with many N atoms since pyridine is a carbon source. The introduction of N reduces the energy band gap and energy barrier to increase the electron carriers of the carbon material, improving the electronic conductivity.<sup>[43–45]</sup> Moreover, the thickness of the carbon layer encapsulating the Si surface can be regulated, indicating the potential for customizable structure. The average thickness of the carbon coating on the Si surface decreases with increasing pyrolysis temperature, reducing from 14.75 nm in PySi/C8 to 6.19 nm in PySi/C11 (Figure S17, Supporting Information). The high pyrolysis temperature satisfies the nucleation conditions of the gas-phase carbon source and promotes easier self-nucleation. In addition, other pyrolysis conditions can modulate the carbon structure. By elevating the injection rate to 10 ml min<sup>-1</sup>, we obtained PySi/C11 with a thinner carbon layer. Our focus here lies on

investigating the impact of temperature conditions on the structure of PySi/Cs. Therefore, we maintained a fixed injection rate throughout our experiments.

Subsequently, the changes in functional groups and chemical bonds of different PySi/Cs are explored. Fourier-transform infrared spectra demonstrate the presence of nitrogen-containing functional groups and Si-O-Si bonds in PySi/Cs (Figure S18 and Note S3, Supporting Information).<sup>[46–48]</sup> Chemical bonding changes in PySi/Cs upon different preparation temperatures are explored with X-ray absorption near edge structure (XANES) and X-ray photoelectron spectroscopy (XPS). The Si  $L_{2,3}$ -edge spectra contain five characteristic peaks (Figure 2e), of which A and B are a set of double-split characteristic peaks located at 103–103.7 eV, corresponding to the Si crystalline phase.<sup>[49,50]</sup> The double splitting peaks indicate the appearance of spin-orbiting doublets of Si  $2p_{3/2}$  and Si  $2p_{1/2}$ . Another set of split peaks (C and D) located at 106.2–106.8 eV corresponds to the electron leap of Si atoms in silicon oxide.<sup>[51,52]</sup> The feature E gradually shifts toward lower energy as the temperature rises (PySi/C8 at 108.8 eV to PySi/C11 at 108.4 eV), indicating the formation of more amorphous components.<sup>[51]</sup> The high-resolution Si  $2p$  XPS spectra are fitted into four peaks located at 99.6, 100.2, 102.2, and 103.9 eV, corresponding to the Si  $2p_{3/2}$ , Si  $2p_{1/2}$ , Si<sup>2+</sup>, and Si<sup>4+</sup> components, respectively (Figure 2h).<sup>[53,54]</sup> The oxidation degree of Si on the PySi/C surface gradually increases as the preparation temperature rises, with the surface silicon oxide ratio increasing from 61.8% of PySi/C8 to 75.4% of PySi/C11 (Figure S19a, Supporting Information).

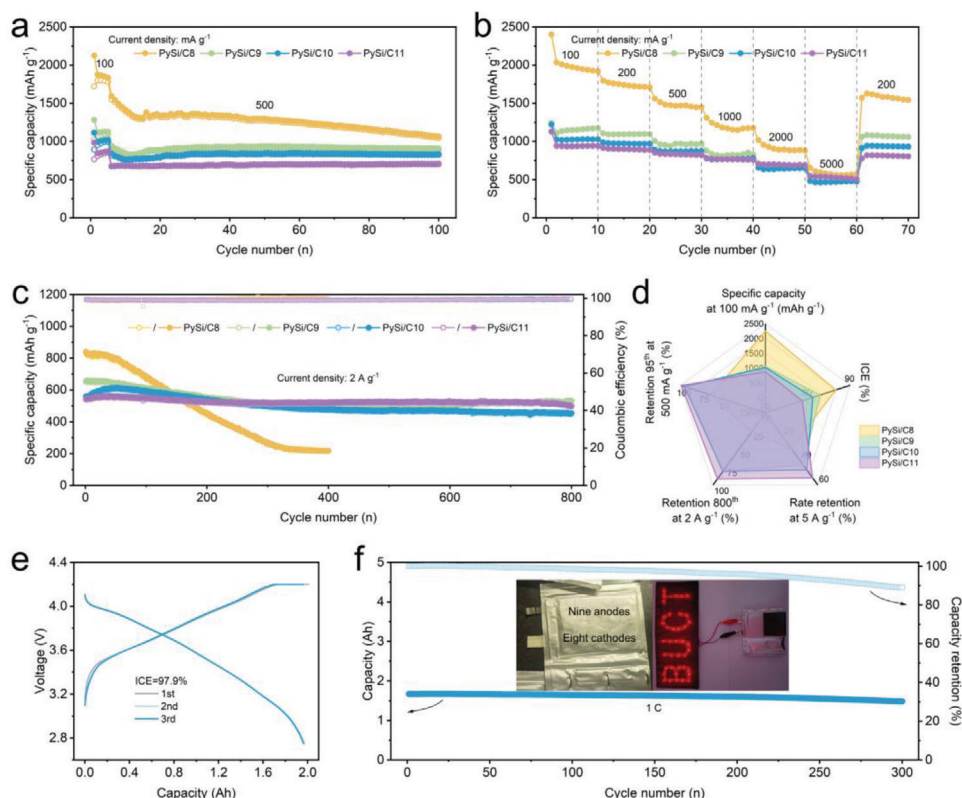
In C-K edge spectra (Figure 2f), the obvious feature  $\pi^*$  at 285.9 eV corresponds to the  $sp^2$  bonding configuration.<sup>[55]</sup> The feature  $\sigma^*$  at 293 eV is attributed to the exciton transition from the core level to the  $\sigma^*$  state. In contrast, the feature A at 288.5 eV between  $\pi^*$  and  $\sigma^*$  bands corresponds to the structural change resulting from the increase of the  $sp^3$ -like structure and C-H at the edge or defect site.<sup>[36,55]</sup> With the increase in pyrolysis temperature, the decrease in the intensity of feature A indicates a reduction in the defect content of the carbon component in PySi/C, which is in full accordance with the TEM results. Furthermore, given that the carbon is doped with nitrogen atoms, the impact of high temperature on the oxidation of the carbon component can be assessed by examining alterations in nitrogen bonding. As shown in N-K edge spectra (Figure 2g), feature A (398.5 eV) and feature B (401.4 eV) correspond to the pyridine ring and oxidized pyridine structures in PySi/Cs, respectively.<sup>[56]</sup> Rising temperatures causes a decrease in the pyridine-N structure and an increase in the degree of oxidation. Meanwhile, the XPS results of C1s and N1s are consistent with the corresponding XANES (Figure S19b, c and Note S4, Supporting Information). The changes in Si and N elements reveal that the pyrolysis temperature should not be too high.

The effect of preparation temperature on PySi/Cs is summarized by analyzing the regularity of carbon formation, material morphology, and structural changes. The carbon content rises, defects and doped atoms reduce, the crystallinity of the carbon component increases, and oxidation increases in accordance with the temperature rise. Except for the graphitic carbon coating on the Si surface, pyrolytic carbon with an onion-like buffer structure is also introduced to build a dual-model structure. The role of this structure in the PySi/Cs' electrochem-

ical performance needs to be comprehensively evaluated and compared.

The cyclic voltammetry curves and corresponding galvanostatic charge and discharge (GCD) curves reveal an increased participation of dual-model carbon in the lithiation reaction with rising temperatures (Figure S20 and Note S5, Supporting Information). The reversible capacities of the four PySi/Cs at 100 mA  $g^{-1}$  are 2236.8, 1125.7, 1023.0, and 883.6 mAh  $g^{-1}$ , corresponding to initial coulombic efficiency (ICE) of 86.6, 82.9, 81.2, and 78.7%, respectively. The PySi/Cs' capacities were compared to the predicted capacities calculated based on Si and pyrolytic carbon (Figure S21 and Table S1, Supporting Information). The comparison reveals that the actual capacity of PySi/Cs is generally in line with the expected capacity, indicating the dual-model carbon structure rarely hinders the lithiation process of Si. Moreover, the PySi/Cs cycle 95 times at 500 mA  $g^{-1}$  after wetting the anodes (100 mA  $g^{-1}$  for 5 times), with capacity retention of 67.4, 97.8, 100.5, and 104.2%, respectively (Figure 3a). The single protection of onion-like carbon coating in PySi/C8 improves the cyclic stability to some extent compared to pristine Si NPs (Figure S22a, Supporting Information). In contrast, the dual-model carbon structure endows PySi/C with stable cycling performance. The rate performance test is carried out at current densities ranging from 100 to 5000 mA  $g^{-1}$  (Figure 3b). While PySi/C8 has the highest capacity at 100 mA  $g^{-1}$ , its capacity retention is only 29.7% at 5000 mA  $g^{-1}$ . PySi/C9, PySi/C10, and PySi/C11 have capacity retention rates of 43.4, 47.4, and 54.1%, respectively. Dual-model carbon constructs a conductivity network in PySi/Cs, ensuring higher electronic conductivity and delivering superior rate performance. An important phenomenon is that despite the obvious difference in Si content between PySi/Cs, their capacities are similar at high current density. The intense polarization experienced at a high current density hampers the passage of Li<sup>+</sup> through the carbon coating, thus affecting the involvement of Si.<sup>[57]</sup> PySi/C11 has a thinner carbon coating, allowing more Si to participate in the lithiation reaction and resulting in higher capacity. Long cycle performances of PySi/Cs are further evaluated at a current density of 2 A  $g^{-1}$  for 800 cycles (Figure 3c). The capacity of PySi/C8 decays rapidly, and its capacity retention rate is only 26.0% after 400 cycles. In contrast, PySi/Cs containing dual-model carbon exhibit more stable cycling performances after 800 cycles, with capacity retention rates of 80.8, 81.2, and 91.5% for PySi/C9, PySi/C10, and PySi/C11, respectively. What's more, PySi/C demonstrates superior comprehensive performance compared to Si-based anodes fabricated on a kilogram scale, as reported in recent years (Figure S22b and Table S2 in Supporting Information).

In the above analysis, the carbon component in PySi/Cs features an onion-like graphitic carbon structure. We further explore the effect of the crystallinity of the carbon component on the electrochemical performance (Figure S23 and Note S6, Supporting Information). The result proves that the high crystallinity graphitic carbon layer is beneficial to maintaining the cycling stability. What's more, the electrochemical performances of four PySi/Cs are comprehensively evaluated in terms of ICE, specific capacity, cycle stability, and rate performance (Figure 3d). The results demonstrate that the dual-model structure plays an important role in improving the electrochemical performance but simultaneously reduces ICE and capacity. Therefore, it is



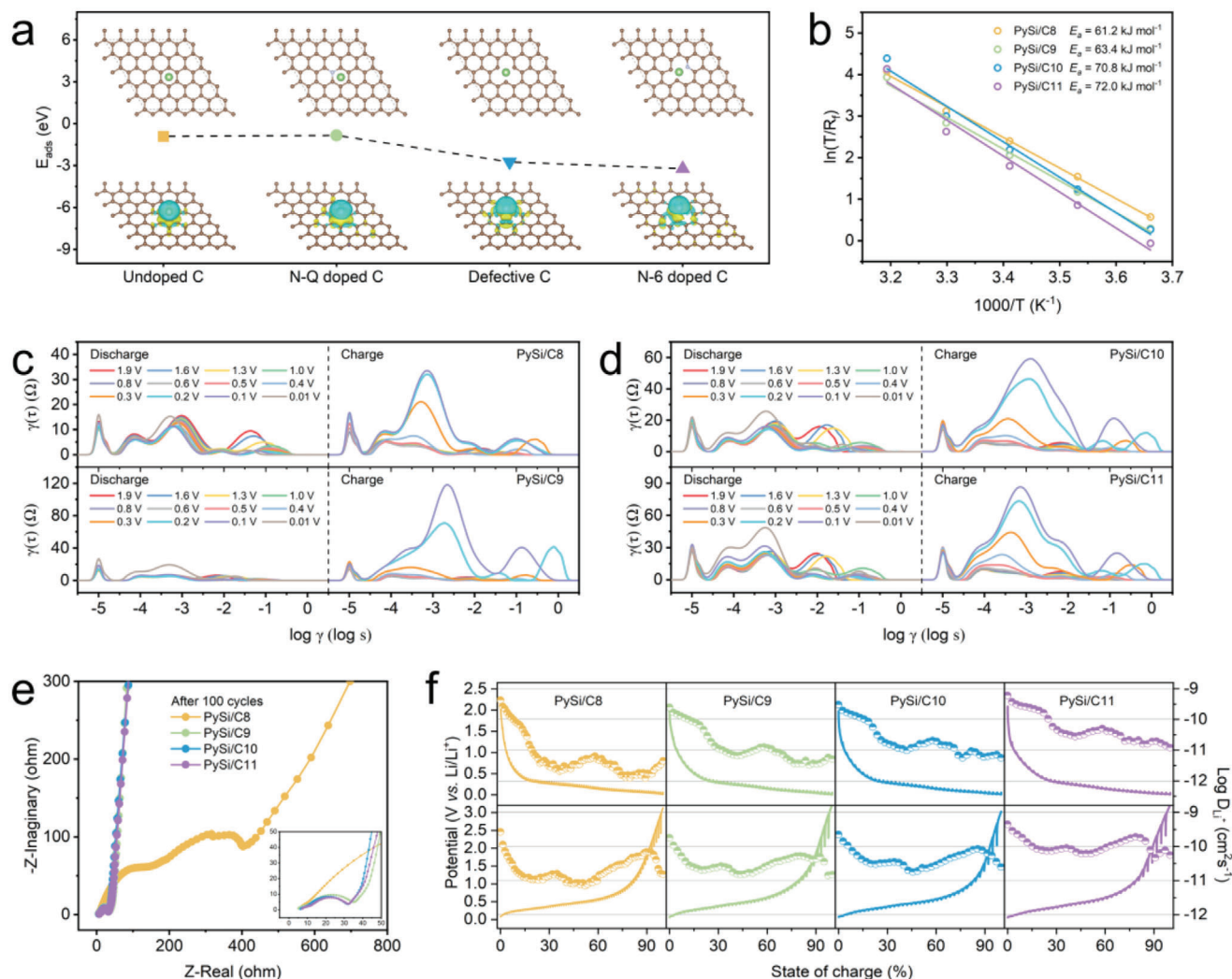
**Figure 3.** a) Short cycle performances, b) rate performances, c) long cycle performances, and d) electrochemical performance comparing image of PySi/Cs. e) Initial GCD curves of 30%PySi/C10+70%Graphite || NCM811 pouch cell at 0.1C. f) Cycle performance of pouch cell at 1C, and embedded images are photos of pouch cell.

possible to selectively prepare PySi/C with suitable properties for different situations. Compared to other PySi/Cs, PySi/C10 has more comprehensive competitiveness in terms of physical properties and electrochemical performances. Therefore, PySi/C10 is assembled into a single pouch full cell by pairing with  $\text{LiFePO}_4$ . The full cell delivers an excellent ICE of 93.9% with a high reversible capacity of  $163.3 \text{ mAh g}^{-1}$  at 0.1 C ( $1 \text{ C} = 170 \text{ mA g}^{-1}$ ) (Figure S24a, Supporting Information). The full cell capacity remains at  $140.7 \text{ mAh g}^{-1}$  with a capacity retention of 97.8% at 0.5 C after 100 cycles, exhibiting excellent cycling stability (Figure S24b, Supporting Information). More importantly, PySi/C10 is mixed with graphite (mass ratio 3:7) and then paired with  $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$  (NCM811), assembling into a pouch-type full battery with a rated capacity of 2 Ah to assess the potential for practical applications. Figures S25, S26, and Table S3 (Supporting Information) illustrate the electrochemical characteristics of the graphite and the fabrication procedure for pouch cells, respectively. After the formation process, the reversible capacity of the pouch cell at 0.1 C ( $1 \text{ C} = 200 \text{ mA g}^{-1}$ ) is 1.97 Ah with an ICE of 97.9% (Figure 3e). The pouch cell delivers stable performance, maintaining a capacity of 1.49 Ah after 300 cycles at 1 C with a retention of 89.0% and easily lighting up an array of 40 LEDs (Figure 3f). The pouch cell conducts an energy density of  $248.2 \text{ Wh kg}^{-1}$  (calculated using Equation S1, Supporting Information), close to the designed  $250 \text{ Wh kg}^{-1}$ . Such superior electrochemical performances of PySi/C full batteries demonstrate application potential in the field of Si/C anodes.

Figure 4a displays the optimized model along with the associated adsorption energy ( $E_{\text{ads}}$ ) concerning  $\text{Li}^+$  and the corresponding differential charge density. Based on the calculation results, the most charge transfer exists for the  $\text{Li}^+$  adsorption reaction of pyridinic-N (N-6) doped C, and its  $E_{\text{ads}}$  of  $-3.22 \text{ eV}$  is lower than those of undoped C ( $-0.91 \text{ eV}$ ), graphitic-N (N-Q) doped C ( $-0.84 \text{ eV}$ ), and defective C ( $-2.74 \text{ eV}$ ). In conjunction with the findings, the defects and the N-6 doping concentration in the carbon structures decrease with rising pyrolysis temperature. We deduce that the adsorption of  $\text{Li}^+$  becomes more difficult for PySi/C as the pyrolysis temperature increases. Electrochemical kinetics is assessed to thoroughly explore the variations in the electrochemical performance of PySi/Cs via electrochemical impedance spectroscopy (EIS). Four anodes are tested from 0 to  $40^\circ\text{C}$  with an increment step of  $10^\circ\text{C}$  after 10 cycles, and the corresponding EIS curves are displayed in Figure S27 (Supporting Information). The activation energy ( $E_a$ ) for  $\text{Li}^+$  transport across the surface of PySi/C is calculated using Equation S2 (Supporting Information), resulting in values of  $61.2 \text{ kJ mol}^{-1}$  for PySi/C8,  $63.4 \text{ kJ mol}^{-1}$  for PySi/C9,  $70.8 \text{ kJ mol}^{-1}$  for PySi/C10, and  $72.0 \text{ kJ mol}^{-1}$  for PySi/C11 (Figure 4b). Raising the pyrolysis temperature enhances crystallinity and reduces the content of defects and dopants in onion-like carbon components, resulting in a decrease in active sites for  $\text{Li}^+$ . These results are in accordance with theoretical calculations.

The EIS results are analyzed using the distribution of relaxation time (DRT) transform to distinguish resistance changes



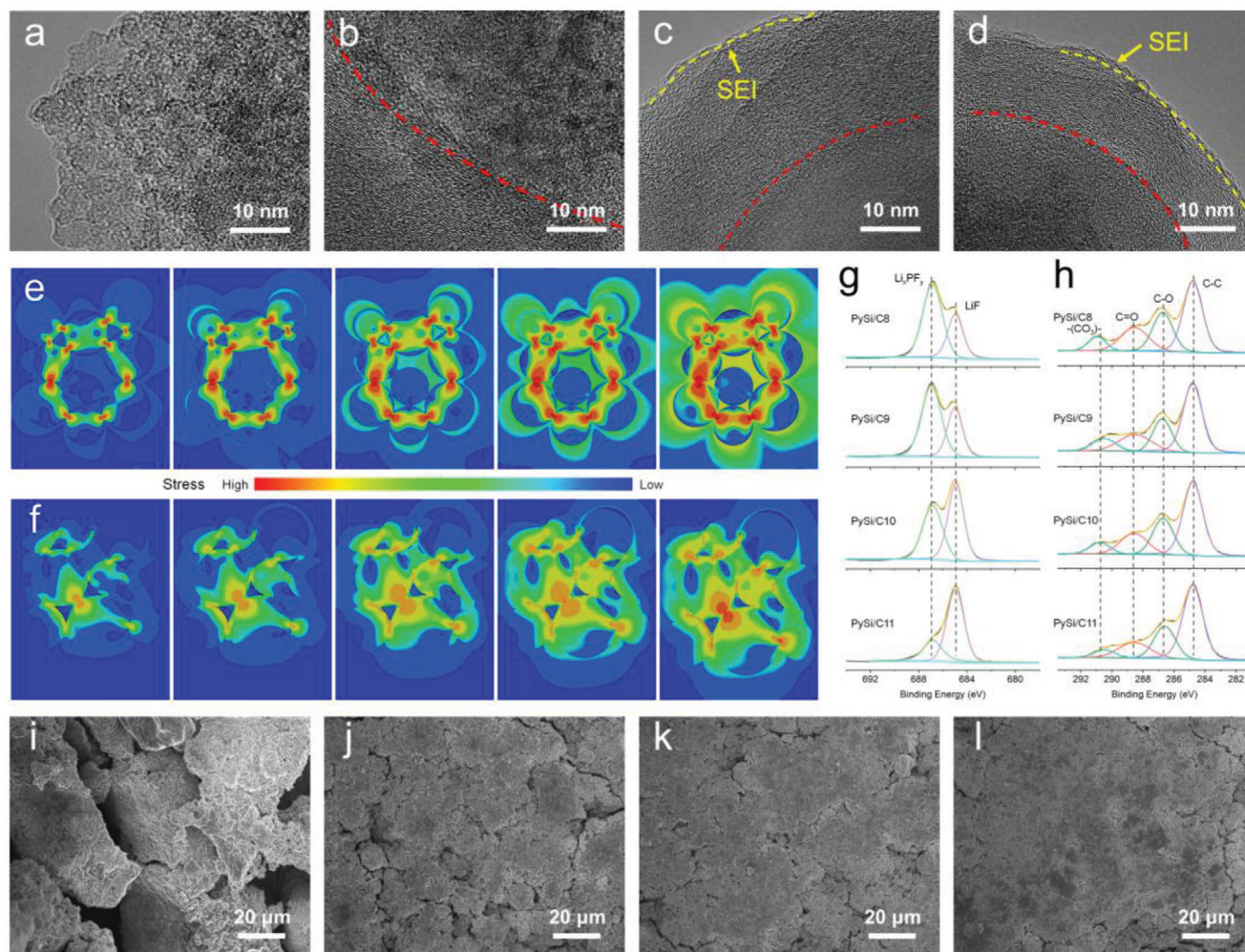


**Figure 4.** a) DFT calculation of  $\text{Li}^+$ -adsorption in different carbon model and the corresponding differential charge density between C atoms and  $\text{Li}^+$  (yellow and cyan isosurfaces indicate the accumulation and depletion of the electronic charge density, respectively, with units set to  $0.002 \text{ e } \text{\AA}^{-3}$ ). b) Activation energy in  $\text{Li}^+$  transporting. c, d) DRT curves of PySi/Cs at different charged states. e) Nyquist plots of PySi/Cs after 100 cycles. f) GITT curves of PySi/Cs.

during electrochemical process (Figure S28 and Note S7, Supporting Information).<sup>[58–60]</sup> The higher SEI resistance ( $R_f$ ) contributed by different components with increasing preparation temperature at 2 V indicates that more dual-model carbon structures are involved in the electrochemical reaction. In-situ EIS measurements and DRT analysis on half-cells are carried out to compare resistance distribution (Figure 4c,d; Figure S29, Supporting Information). The anodes are cycled 10 times at  $100 \text{ mA g}^{-1}$  prior to the in-situ EIS test to eliminate the impacts of SEI film formation. The noteworthy phenomenon is that the  $R_f$  reaches its maximum value after delithiation at 0.1 V. The contraction of Si NPs during the delithiation process tends to close the cracks in the SEI film. This closure leads to an impedance rise, as  $\text{Li}^+$  is hindered from quickly accessing the reaction site through the cracks.<sup>[61]</sup> This result indicates that the presence of pyrolytic carbon causes the anode to develop more and denser SEI films, which are elastic and help mitigate the volume ex-

pansion of Si during the cycling process. What's more, the decrease in charge transfer resistance ( $R_{ct}$ ) suggests that the high crystallinity carbon composition promotes improved kinetics. Although the high-frequency section of the EIS curves for all anodes at 10 cycles maintains a small semicircular diameter (Figure S30, Supporting Information), the semicircular diameter of PySi/C8 increases destructively after 100 cycles (Figure 4e). This suggests that the unstable growth of SEI eventually leads to enormous reaction resistance. In contrast, other PySi/Cs still maintain small resistances, indicating that the dual-model structure ensures the stability of the electrodes and facilitates electron transport.

Subsequently, the galvanostatic intermittent titration technique (GITT) is used to study the lithium-ion migration mechanism of different PySi/C, and the corresponding lithium-ion diffusion coefficient ( $D_{\text{Li}^+}$ ) is calculated in Figure 4f (according to the Equation S3, Supporting Information). The average



**Figure 5.** HRTEM images after 50 cycles of a) PySi/C8, b) PySi/C9, c) PySi/C10, and d) PySi/C11. Stress distribution simulation at different lithiation states e) PySi/C with single coating and f) PySi/C with dual model. XPS g) F 1s and h) C 1s spectra of PySi/Cs. Top-section SEM images after 100 cycles of i) PySi/C8, j) PySi/C9, k) PySi/C10, and l) PySi/C11.

$D_{\text{Li}^+}$  of PySi/C11 anode is calculated as  $7.09 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ , which is higher than the  $D_{\text{Li}^+}$  for PySi/C8 ( $1.05 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ ), PySi/C9 ( $2.01 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ ), and PySi/C10 anode ( $2.9 \times 10^{-11} \text{ cm}^2 \text{ s}^{-1}$ ). The GITT results indicate that changes in the graphitization degree and size of the carbon components will promote the diffusion of  $\text{Li}^+$  in the pyrolysis temperature range.

Structural characterization shows that the carbon coating on the Si surface decreases with increasing temperature, while the amount of pyrolytic carbon increases. Therefore, we further investigated which carbon structure plays a more important role in performance. Si NPs were mixed with pyrolytic carbon obtained at 1000 °C according to the Si/C ratio of PySi/C10. Its capacity retention of 67.9% is much lower than that of PySi/C10 (Figure S31, Supporting Information), suggesting that pyrolytic carbon alone is difficult to ensure structural stability. We also compared the electrochemical properties between PySi/Cs and single-model Si/C of similar capacity (Table S4, Supporting Information). PySi/Cs with dual model carbon structure exhibit superior cycling stability. The indispensable role of each carbon com-

ponent, working synergistically, ensures the stable cycling of the silicon-based anode.

The structural integration of PySi/Cs anodes are examined after cycling. No onion-like carbon structure is observed in PySi/C8 from HRTEM after 50 cycles (Figure 5a), and Si NPs have completely lost their initial spherical shape after repeated expansion and contraction. In contrast, PySi/C9, PySi/C10, and PySi/C11 maintain the original structure (Figure 5b–d), in which Si NPs are completely encapsulated in the graphitic carbon coating, indicating the effectiveness of dual model on stable structure to ensure superior electrochemical performance. In addition, the integral core-shell arrangement allows for the formation of a dense and stable SEI film on the surface of the carbon coating (outside the yellow dotted line). The finite element analysis was carried out to simulate the lithiation expansion stresses of the two structured electrodes, where Figure S32 (Supporting Information) corresponds to the initial models of single-coated PySi/C and the dual-model structure PySi/C, respectively. The variation in stress distribution for different lithiation states of the two types of electrodes are shown in Figure 5e,f, respectively. Single-coated



PySi/C is subjected to excessive expansion stresses after lithiation. Especially in the particle contact location, it generates an enormous stress concentration. Severe stress will destroy the carbon coating structure and cause the particles to migrate, resulting in large cracks in the electrode.<sup>[11]</sup> When the dual-model structure is introduced, the pyrolytic carbon serves as a stress relieving medium to absorb the stress of the Si/C particles and reduce the stress concentration between the particles. The two carbon compositions synergistically minimize the overall expansion, ensuring structural integration and endowing superior electrochemical performances.

The XPS analysis of the anode surface after cycling highlights the significance of the dual-model structure in enhancing cyclic stability (Figure 5g,h; Figure S33, Supporting Information). In the F 1s spectra, peaks at 685.1 eV and 686.9 eV correspond to LiF and  $\text{Li}_x\text{PF}_y$ , respectively.<sup>[62]</sup> In the C 1s spectra, four characteristic peaks are observed, representing C–C (284.8 eV), C–O (286.7 eV), C=O (288.5 eV), and carbonate (290.8 eV).<sup>[63]</sup> With increasing pyrolysis temperature, the proportion of LiF in the SEI on the PySi/Cs electrode surface gradually increases, while unstable organic components decrease, indicating the formation of a more robust SEI.<sup>[64,65]</sup> Moreover, the rising proportion of C–C components reflects the improved structural stability of PySi/Cs. The morphological changes of PySi/Cs electrodes after cycling are then compared based on the results of the finite element analysis. All anodes have a dense surface morphology before cycling, as shown in Figure S34 (Supporting Information). However, visible differences are identified among them after 100 cycles (Figure 5i–l). The PySi/C8 electrode exhibits considerable cracks, and the active material has been stripped from the current collector. The other PySi/Cs electrodes retain stable structures, and the cracks gradually decrease with the increase in dual-model structure. The delicate design endows PySi/Cs with stable SEI films and structures. The unique onion-like structure of carbon coating and pyrolytic carbon reduces the expansion stress caused by Si lithiation through interlayer slip and avoids stress concentration.<sup>[24,66]</sup> This dual-model structure efficiently maintains electrode structural integrity, prevents crack generation and propagation, and ensures the cyclic stability of the electrode.

### 3. Conclusion

In summary, PySi/Cs with a dual-model structure are successfully prepared by a large-scale synthesis strategy. We also design a new injection pyrolysis equipment with superior industrial compatibility. The method is based on deposition theory to realize the controllable conformation of two carbon components simultaneously, which is distinctly different from the existing multistage structure strategies. We reveal the mechanism of formation of the dual-model carbon structure and detail the coupling of the two carbon nucleation laws as well as the controllable changes in morphology and crystallinity. An in-depth study is conducted on the impact of the dual-model carbon structure on the ICE, cycling performance, electrochemical kinetics, and electrode structure stability of PySi/C. As a proof, the dual-model structure ensures the stability of the structure by reducing the stress concentration at the contact interface through the transferred stress. Meanwhile, the onion-like graphitic carbon also provide an efficient medium for lithium-ion diffusion, enabling rapid transfer

of lithium ions and electrons. This work offers significant novel perspectives and valuable concepts for the preparation of anode materials, which can contribute to the development of the energy storage device field.

### Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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### Conflict of Interest

The authors declare no conflict of interest.

### Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article

### Keywords

dual-model carbon structures, injection pyrolysis, kilogram scales, silicon-carbon composites

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- [1] H. Li, H. Li, Y. Lai, Z. Yang, Q. Yang, Y. Liu, Z. Zheng, Y. Liu, Y. Sun, B. Zhong, Z. Wu, X. Guo, *Adv. Energy Mater.* **2022**, 12, 2102181.
- [2] M. Ko, S. Chae, J. Ma, N. Kim, H.-W. Lee, Y. Cui, J. Cho, *Nat. Energy* **2016**, 1, 1.
- [3] S. Chae, S. H. Choi, N. Kim, J. Sung, J. Cho, *Angew. Chem., Int. Ed.* **2020**, 59, 110.
- [4] P. Li, H. Kim, S.-T. Myung, Y.-K. Sun, *Energy Stor. Mater.* **2021**, 35, 550.
- [5] K. Ogata, E. Salager, C. J. Kerr, A. E. Fraser, C. Ducati, A. J. Morris, S. Hofmann, C. P. Grey, *Nat. Commun.* **2014**, 5, 3217.
- [6] S. Pan, J. Han, Y. Wang, Z. Li, F. Chen, Y. Guo, Z. Han, K. Xiao, Z. Yu, M. Yu, S. Wu, D. W. Wang, Q. H. Yang, *Adv. Mater.* **2022**, 34, 2203617.
- [7] Q. Shi, J. Zhou, S. Ullah, X. Yang, K. Tokarska, B. Trzebicka, H. Q. Ta, M. H. Rummeli, *Energy Stor. Mater.* **2021**, 34, 735.
- [8] B. Wang, X. Li, B. Luo, L. Hao, M. Zhou, X. Zhang, Z. Fan, L. Zhi, *Adv. Mater.* **2015**, 27, 1526.
- [9] D. Wang, C. Zhou, B. Cao, Y. Xu, D. Zhang, A. Li, J. Zhou, Z. Ma, X. Chen, H. Song, *Energy Stor. Mater.* **2020**, 24, 312.
- [10] S.-B. Son, L. Cao, T. Yoon, A. Cresce, S. E. Hafner, J. Liu, M. Groner, K. Xu, C. Ban, *Adv. Sci.* **2019**, 6, 1801007.
- [11] J.-B. Fang, Y.-Q. Cao, S.-Z. Chang, F.-R. Teng, D. Wu, A.-D. Li, *Adv. Funct. Mater.* **2022**, 32, 2109682.

- [12] J. Zhang, D. Wang, R. Yuan, X. Li, J. Li, Z. Jiang, A. Li, X. Chen, H. Song, *Small*. **2023**, 19, 2207167.
- [13] J. Shi, L. Zu, H. Gao, G. Hu, Q. Zhang, *Adv. Funct. Mater.* **2020**, 30, 2002980.
- [14] L. Zhang, Q. Huang, X. Liao, Y. Dou, P. Liu, M. Al-Mamun, Y. Wang, S. Zhang, S. Zhao, D. Wang, G. Meng, H. Zhao, *Energy Environ. Sci.* **2021**, 14, 3502.
- [15] C. Xu, L. Shen, W. Zhang, Y. Huang, Z. Sun, G. Zhao, Y. Lin, Q. Zhang, Z. Huang, J. Li, *Energy Stor. Mater.* **2023**, 56, 319.
- [16] M. H. Parekh, A. D. Sediako, A. Naseri, M. J. Thomson, V. G. Pol, *Adv. Energy Mater.* **2020**, 10, 1902799.
- [17] H. Li, X. Li, D. Wang, S. Zhang, W. Xu, L. N. Zhu, L. Zhi, *Nanoscale*. **2021**, 13, 2820.
- [18] H. Su, X. Li, C. Liu, Y. Shang, H. Liu, *Chem. Eng. J.* **2023**, 451, 138394.
- [19] Q. Xu, J.-K. Sun, J.-Y. Li, Y.-X. Yin, Y.-G. Guo, *Energy Stor. Mater.* **2018**, 12, 54.
- [20] Q. Xu, J.-Y. Li, J.-K. Sun, Y.-X. Yin, L.-J. Wan, Y.-G. Guo, *Adv. Energy Mater.* **2017**, 7, 1601481.
- [21] J. Li, Y. Huang, W. Huang, J. Tao, F. Lv, R. Ye, Y. Lin, Y. Y. Li, Z. Huang, J. Lu, *Small*. **2021**, 17, 2006373.
- [22] S. Zhu, J. Zhou, Y. Guan, W. Cai, Y. Zhao, Y. Zhu, L. Zhu, Y. Zhu, Y. Qian, *Small*. **2018**, 14, 1802457.
- [23] A. Magasinski, P. Dixon, B. Hertzberg, A. Kvit, J. Ayala, G. Yushin, *Nat. Mater.* **2010**, 9, 353.
- [24] I. H. Son, J. Hwan Park, S. Kwon, S. Park, M. H. Rummeli, A. Bachmatiuk, H. J. Song, J. Ku, J. W. Choi, J. M. Choi, S. G. Doo, H. Chang, *Nat. Commun.* **2015**, 6, 7393.
- [25] C. Qi, S. Li, Z. Yang, Z. Xiao, L. Zhao, F. Yang, G. Ning, X. Ma, C. Wang, J. Xu, J. Gao, *Carbon*. **2022**, 186, 530.
- [26] G. Huang, J. Han, Z. Lu, D. Wei, H. Kashani, K. Watanabe, M. Chen, *ACS Nano*. **2020**, 14, 4374.
- [27] X. Zhang, D. Wang, X. Qiu, Y. Ma, D. Kong, K. Mullen, X. Li, L. Zhi, *Nat. Commun.* **2020**, 11, 3826.
- [28] B. Cao, H. Liu, Z. Xing, Y. Lei, H. Song, X. Chen, J. Zhou, Z. Ma, *ACS Sustainable Chem. Eng.* **2015**, 3, 1786.
- [29] X. Hong, T.-C. Zhang, L.-D. Zhang, F. Qi, *Chin. J. Chem Phys.* **2009**, 22, 204.
- [30] A. Becker, K. J. Huttinger, *Carbon*. **1997**, 36, 201.
- [31] B. Banerjee, T. Hirt, P. Walker, *Nature*. **1961**, 192, 450.
- [32] F. Quli, P. Thrower, L. Radovic, *Carbon*. **1998**, 36, 1623.
- [33] J. Lahaye, F. Ehrburger-Dolle, *Carbon*. **1994**, 1319.
- [34] Y. An, Y. Tian, Y. Zhang, C. Wei, L. Tan, C. Zhang, N. Cui, S. Xiong, J. Feng, Y. Qian, *ACS Nano*. **2020**, 14, 17574.
- [35] J. Lu, J. Liu, X. Gong, S. Pang, C. Zhou, H. Li, G. Qian, Z. Wang, *Energy Stor. Mater.* **2022**, 46, 594.
- [36] C. Meng, M. Yuan, B. Cao, X. Lin, J. Zhang, A. Li, X. Chen, M. Jia, H. Song, *Carbon*. **2022**, 192, 347.
- [37] S. Zhao, Z. Yang, X. Zhou, K. Sun, *Ceram Int.* **2016**, 42, 9264.
- [38] M. Escobar, M. S. Moreno, R. J. Candal, M. C. Marchi, A. Caso, P. I. Polosecki, G. H. Rubiolo, S. Goyanes, *Appl. Surf. Sci.* **2007**, 254, 251.
- [39] J. Lahaye, G. Prado, J. Donnet, *Carbon*. **1974**, 12, 27.
- [40] N. Kim, H. Cha, S. Chae, T. Lee, Y. Lee, Y. Kim, J. Sung, J. Cho, *Energy Environ. Sci.* **2023**, 16, 2505.
- [41] Y. Li, K. Yan, H.-W. Lee, Z. Lu, N. Liu, Y. Cui, *Nat. Energy*. **2016**, 1, 15029.
- [42] R. Yu, Y. Pan, Y. Jiang, L. Zhou, D. Zhao, G. Van Tendeloo, J. Wu, L. Mai, *Adv. Mater.* **2023**, 35, 2306504.
- [43] J. Zhou, Y. Lu, L. Yang, W. Zhu, W. Liu, Y. Yang, K. Liu, *Carbon Energy*. **2022**, 4, 1.
- [44] C. Dong, L. Wu, Y. He, Y. Zhou, X. Sun, W. Du, X. Sun, L. Xu, F. Jiang, *Small*. **2020**, 16, 2004580.
- [45] B.-H. Hou, Y.-Y. Wang, D.-S. Liu, Z.-Y. Gu, X. Feng, H. Fan, T. Zhang, C. Lü, X.-L. Wu, *Adv. Funct. Mater.* **2018**, 28, 1805444.
- [46] R. Yuan, Y. Dong, R. Hou, L. Shang, J. Zhang, S. Zhang, X. Chen, H. Song, *Chem. Eng. J.* **2023**, 454, 140418.
- [47] G. Zhu, F. Zhang, X. Li, W. Luo, L. Li, H. Zhang, L. Wang, Y. Wang, W. Jiang, H. K. Liu, S. X. Dou, J. Yang, *Angew. Chem., Int. Ed.* **2019**, 58, 6669.
- [48] S. Gong, Y. Lee, J. Choi, M. Lee, K. Y. Chung, H. G. Jung, S. Jeong, H. S. Kim, *Small*. **2023**, 19, 2206238.
- [49] V. A. Terekhov, S. Y. Turishchev, V. M. Kashkarov, E. P. Domashevskaya, A. N. Mikhailov, D. I. Tetel'baum, *Phys. E*. **2007**, 38, 16.
- [50] V. A. Terekhov, S. Y. Turishchev, V. M. Kashkarov, E. P. Domashevskaya, A. N. Mikhailov, D. I. Tetel'baum, *J. Surf. Invest.-X-Ray+* **2007**, 1, 55.
- [51] H. M. Tsai, S. C. Ray, C. W. Pao, J. W. Chiou, C. L. Huang, C. H. Du, W. F. Pong, M. H. Tsai, A. Fukano, H. Oyanagi, *J. Appl. Phys.* **2008**, 103, 013704.
- [52] Z. Jiang, R. Yuan, Z. Jiang, J. Zhang, L. Liu, A. Li, X. Chen, H. Song, *Small*. **2023**, 19, 2303294.
- [53] Y. Ma, H. Qu, W. Wang, Y. Yu, X. Zhang, B. Li, L. Wang, *Adv. Funct. Mater.* **2022**, 33, 2211648.
- [54] X. Zhou, Y. Liu, Y. Ren, T. Mu, X. Yin, C. Du, H. Huo, X. Cheng, P. Zuo, G. Yin, *Adv. Funct. Mater.* **2021**, 31, 2101145.
- [55] Y. Dong, S. Zhang, X. Du, S. Hong, S. Zhao, Y. Chen, X. Chen, H. Song, *Adv. Funct. Mater.* **2019**, 29, 1901127.
- [56] A. Vairavamurthy, S. Wang, *Environ. Sci. Technol.* **2002**, 36, 3050.
- [57] Z. Zhang, N. Liao, H. Zhou, W. Xue, *Acta Mater.* **2019**, 178, 173.
- [58] N. Qin, L. Jin, G. Xing, Q. Wu, J. Zheng, C. Zhang, Z. Chen, J. P. Zheng, *Adv. Energy Mater.* **2023**, 13, 2204077.
- [59] Y. Lu, C.-Z. Zhao, J.-Q. Huang, Q. Zhang, *Joule*. **2022**, 6, 1172.
- [60] T. H. Wan, M. Saccoccio, C. Chen, F. Ciucci, *Electrochim. Acta*. **2015**, 184, 483.
- [61] K. Pan, F. Zou, M. Canova, Y. Zhu, J.-H. Kim, *J. Power Sources*. **2020**, 479, 229083.
- [62] J. Tang, F. Wu, X. Dai, J. Zhou, H. Pang, X. Duan, B. Xiao, D. Li, J. Long, *Chem. Eng. J.* **2023**, 452, 139139.
- [63] X. Zhang, S. Weng, G. Yang, Y. Li, H. Li, D. Su, L. Gu, Z. Wang, X. Wang, L. Chen, *Cell Rep. Phys. Sci.* **2021**, 2, 100668.
- [64] J. Chen, X. Fan, Q. Li, H. Yang, M. R. Khoshi, Y. Xu, S. Hwang, L. Chen, X. Ji, C. Yang, H. He, C. Wang, E. Garfunkel, D. Su, O. Borodin, C. Wang, *Nat. Energy*. **2020**, 5, 386.
- [65] J. Wang, Y. Cui, *Nat. Energy*. **2020**, 5, 361.
- [66] X. Han, Z. Zhang, H. Chen, Q. Zhang, S. Chen, Y. Yang, *ACS Appl. Mater. Interfaces*. **2020**, 12, 44840.